PAMELA QUARTSON

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Education

UC Berkeley **May 2026**

Master of Information Management and Systems

- **Relevant Coursework:** Principles & Techniques of Data Science, Building Data Products, Introduction to Data Structures and Analytics, Research Design and Applications for Data and Analysis, Applied Generative AI, Applied Natural Language Processing

Graduate Certificate in Applied Data Science

Ashesi University **May 2021**

BSc, Computer Science

- **Relevant Coursework:** Data Systems & Machine Learning, Data Science, Data Structures & Algorithms, Algorithm Design & Analysis, Computer Organization and Architecture, Networks and Distributed Computing

Skills

- **Languages:** Python, JavaScript/TypeScript, PHP, SQL, R, Java
- **Frameworks & Tools:** Node.js, NestJS, ExpressJS, ReactJS/Redux, Vue, PostgreSQL, MySQL, Google Firebase, MongoDB, Redis
- **Cloud & Infrastructure:** Google Cloud Platform (GCP), Amazon Web Services (AWS), Docker, Kubernetes, DevOps
- **AI/ML:** RAG, GraphRAG, LangChain, ChromaDB, CrewAI, Amazon Bedrock

Work Experience

***Lead Data Engineer* | UC Berkeley Goldman School of Public Policy** **Sep 2025 – Present**

- Lead maintenance and expansion of automated Python ELT pipelines for the Voluntary Registry Offsets Database (VROD), integrating multiple third-party registry APIs to ingest, transform, and centralize 10,000+ carbon offset project records into a unified data store.
- Own end-to-end pipeline reliability: designed schema structures, implemented error handling, automated data validation, and built monitoring logic, reducing manual intervention by 30% and improving data accuracy across all registry sources.
- Continuously refine transformation logic and data models to ensure consistent, clean metrics across heterogeneous data sources, supporting market transparency for researchers and analysts.

***Research Analyst* | UC Berkeley School of Information/ US Treasury Department** **Aug 2025 – Dec 2025**

- Partnered with an interdisciplinary team of researchers and policymakers to validate a decision-theoretic framework converting ML predictions into policy actions, targeting tax credit take-up for 2M+ eligible college students.
- Conducted sensitivity and scenario analyses in R to stress-test model assumptions and quantify how parameter uncertainty propagates into policy recommendations.
- Engineered synthetic data pipelines in R to enable rigorous controlled testing under confidentiality constraints, with no access to real Treasury data.
- Maintained research continuity and adapted workflows during operational disruptions (government shutdown), delivering analyses on schedule under significant ambiguity.
- Translated technical findings for non-technical Treasury stakeholders, bridging academic research and real-world policy application.

***Software Engineer (Full-Stack)* | Street Streams Limited** **Jul 2022 – Jul 2024**

- Designed cloud-native backend infrastructure on GCP (Compute Engine, Cloud SQL, Cloud Run), collaborating with cross-functional teams to build scalable, production-ready systems for the e-commerce platform.
- Built a real-time analytics dashboard aggregating business and user metrics from multiple data sources, reducing leadership decision-making time by 80%, and demonstrating end-to-end ownership from data modeling to visualization.
- Achieved a 25% reduction in API response time through performance optimizations using JavaScript service worker threads and PM2 process manager tool, leading to an improved user experience.
- Provided technical support to merchants and field teams, troubleshooting platform issues and translating user feedback into product improvements.

Projects

Satellite Disaster Image Classification | UC Berkeley School of Information

- Built ML pipeline to automatically classify satellite imagery by disaster type (flooding vs. fire) and damage severity (0–3 scale), supporting faster emergency response resource allocation
- Engineered compact image feature vectors using color, texture, and edge statistics, achieving 99% cross-validated accuracy on disaster type and 0.498 macro F1 on damage severity across 4 imbalanced classes
- Diagnosed and communicated model limitations including an unlearnable minority class and a feature expressiveness ceiling, prov iding actionable recommendations for improving damage severity classification